

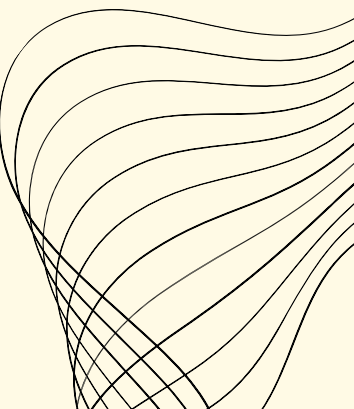


Regional Policy-Making and Analysis

PR. EDUARDO AMARAL HADDAD

Activity 1: "Territorial Development in Morocco: A Spatial Diagnosis Casablanca-Settat provinces

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INTRODUCTION

TERRITORIAL DEVELOPMENT

- Spatial disparities
- More balanced distribution of wealth

32% of the national GDP and 38.9% of industrial establishments

Source: Géographiquement (Climat, Relief, Ecosystème) : rapport de HCP 2020

Tools and Approaches:

- Geographic Information Systems
- GeoDa

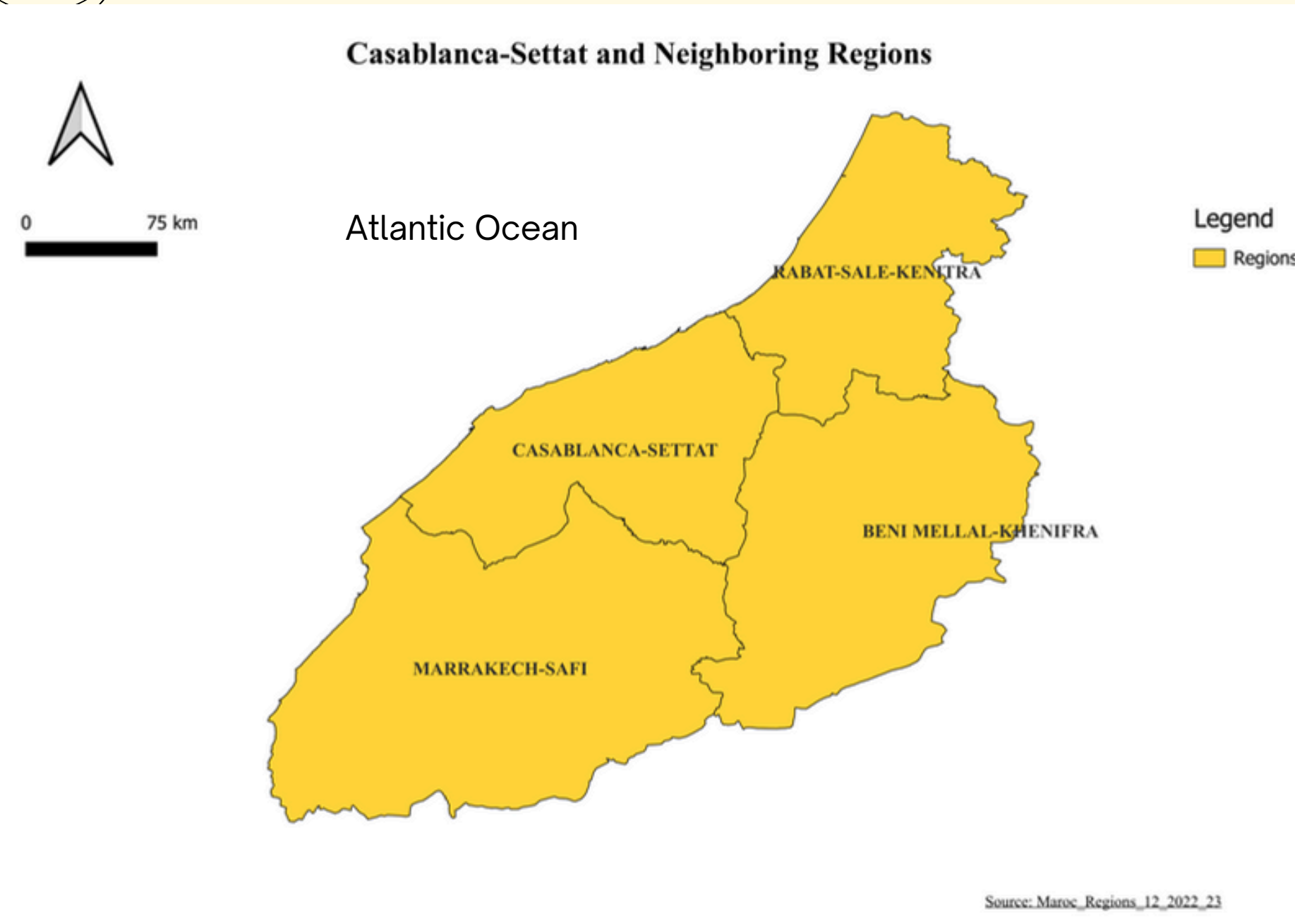
Key indicators:

- Economic performance indicator: total value added and specific sectors
- Demographic indicator: density
- Social indicator: MPI, Education, Health, Total employment

- Formulate strategic recommendations
- Improve territorial balance
- Strengthen the inclusive development dynamics within the Casablanca-Settat region.

I. PRESENTATION OF CASABLANCA IN RELATION TO OTHER REGIONS

GEOGRAPHICALLY



 **Location:** Central-Western Morocco

Total Area: 19,448 km² (2.7% of the national territory)

Source: Géographiquement (Climat, Relief, Ecosystème) : rapport de HCP 2020

DELIMITATION

- **North and West:** The Atlantic Ocean 
- **Northeast:** The Rabat-Salé-Kénitra region
- **East:** The Béni Mellal-Khénifra region
- **South:** The Marrakech-Safi region

Climate and Rainfall of the Region

Region: Atlantic domain

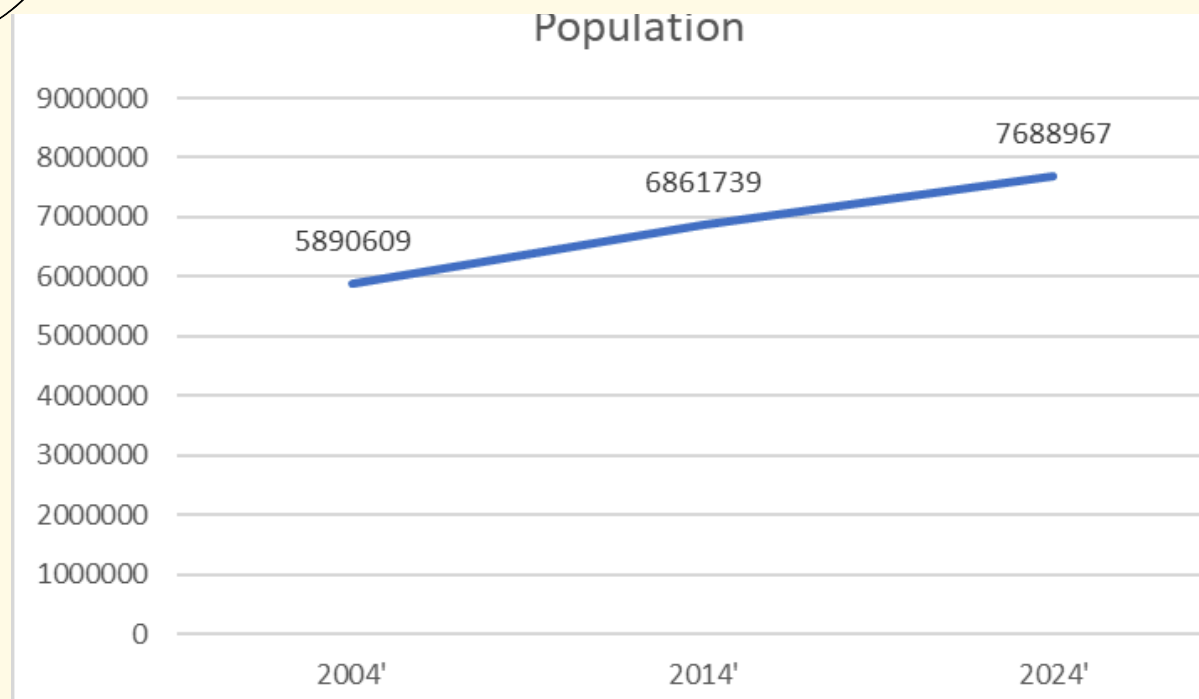
Climate type: Oceanic

Rainfall: 220 to 760 mm

Source: Géographiquement (Climat, Relief, Ecosystème) : rapport de HCP 2020

I. PRESENTATION OF CASABLANCA IN RELATION TO OTHER REGIONS

DEMOGRAPHICALLY



General Population and Housing Census 2014

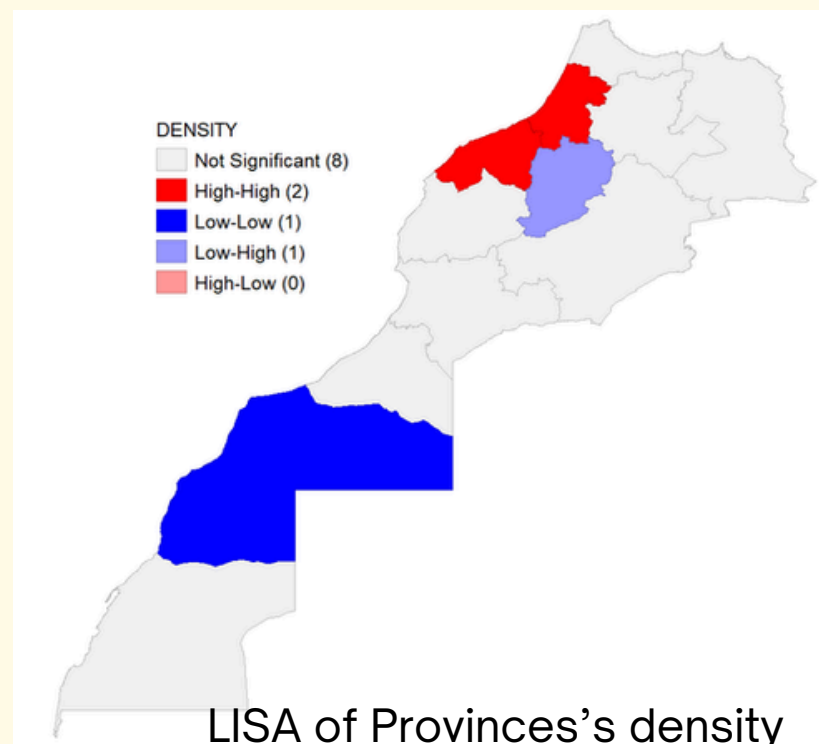
- Legal population (2004): 5,890,609 people
- Legal population (2014): 6,861,739 people
- Legal population (2024): 7 688 967 people
- Relative increase 2014: 16.5%
- Relative increase 2024: 12.1%

SOURCE: GÉOGRAPHIQUÉMENT (CLIMAT, RELIEF, ECOSYSTÈME) : RAPPORT DE HCP 2020

Demographic Concentration of the Casablanca-Settat Region

- Share of national population: 20.3%
- Share of national area: 2.7%
- Population density:
 - Region: 352.8 people/km²
 - National average: 47.6 people/km²

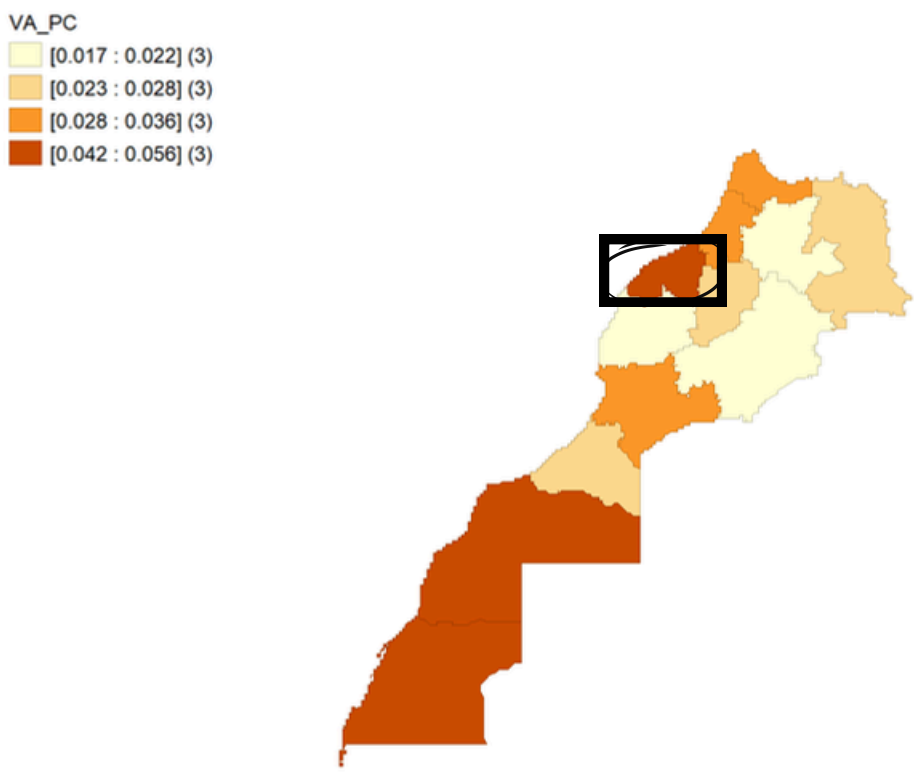
SOURCE: GÉOGRAPHIQUÉMENT (CLIMAT, RELIEF, ECOSYSTÈME) : RAPPORT DE HCP 2020



I. PRESENTATION OF CASABLANCA IN RELATION TO OTHER REGIONS

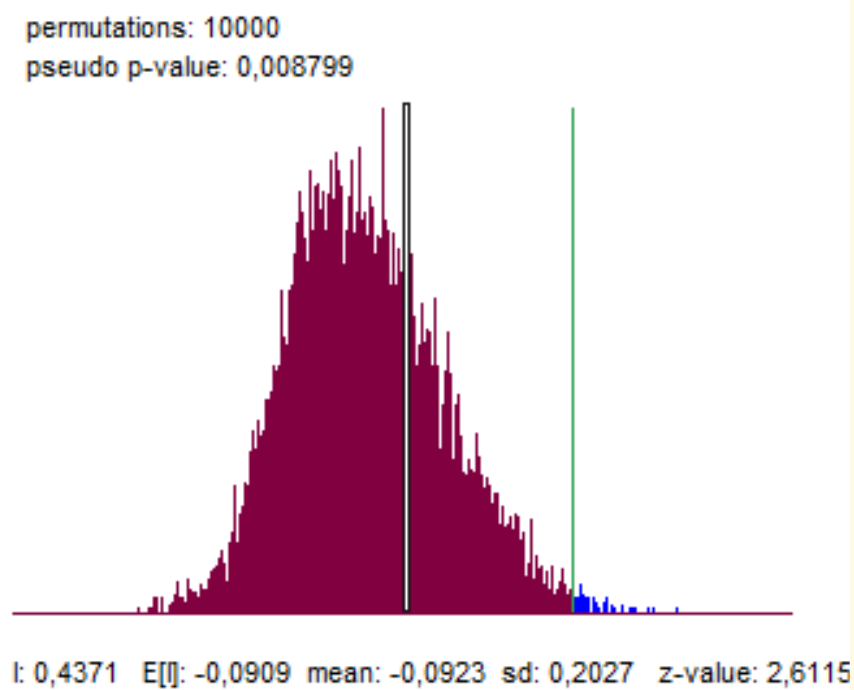
ECONOMICALLY

Added Value per capita of Morocco's Regions



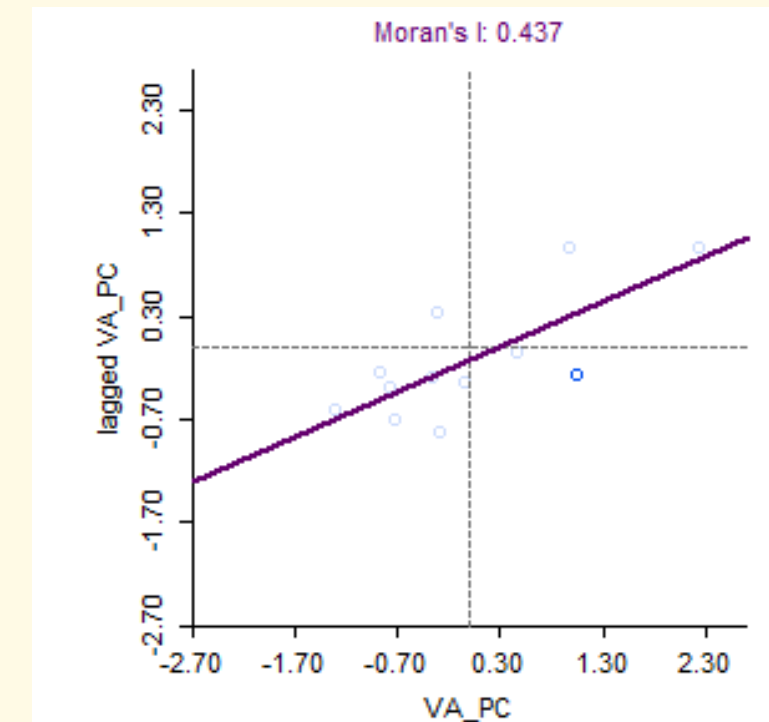
- The map shows Casablanca as a key region in Morocco, with a high added value per capita, highlighting its economic significance

Permutation Test for Moran's I - Statistical Significance of Spatial Autocorrelation



- The permutation test confirms the statistical significance of Moran's I, revealing a meaningful spatial pattern of added value across regions.

Moran's I - Spatial Autocorrelation of Added Value per Capita (VA_PC)



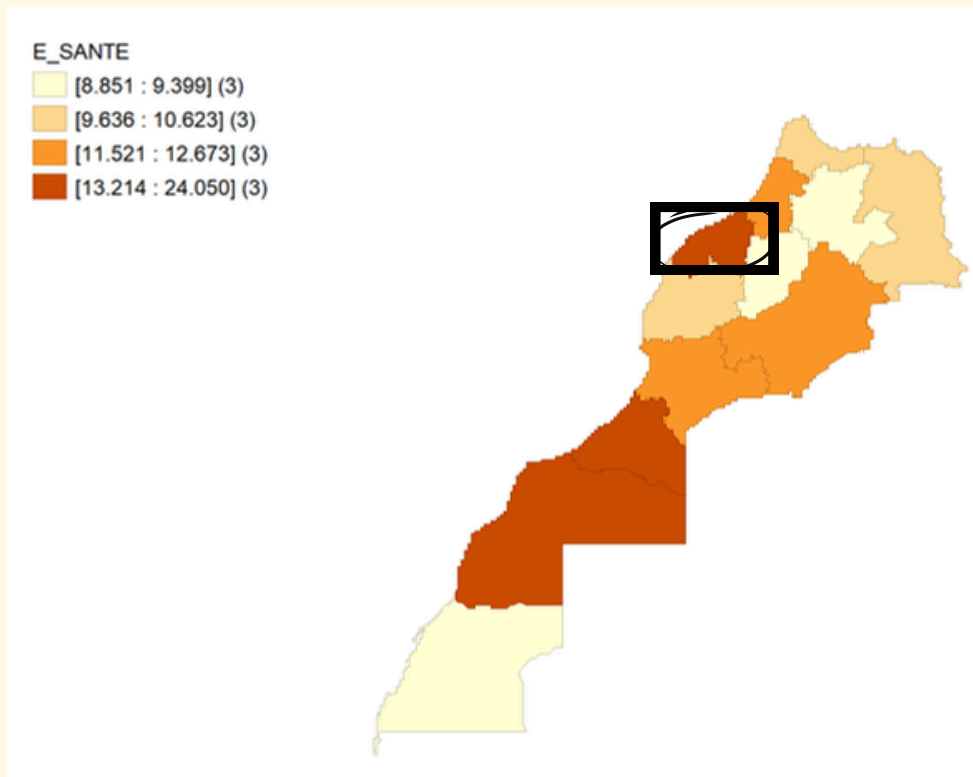
The Moran's I value of 0.437 shows positive spatial correlation, highlighting clustered high and low added value regions, with Casablanca-Settat having higher added value per capita than its neighbors.

I. PRESENTATION OF CASABLANCA IN RELATION TO OTHER REGIONS

O SOCIAL LEVEL

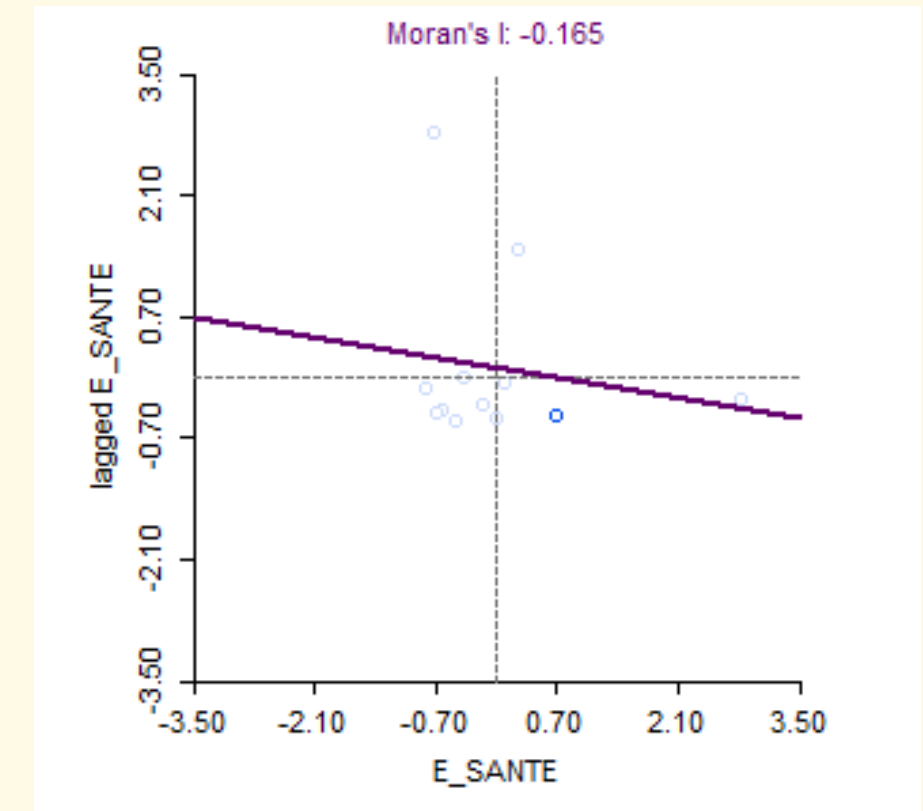
HEALTH

Decomposition of Poverty by Source of Deprivation - Health Dimension (2014)

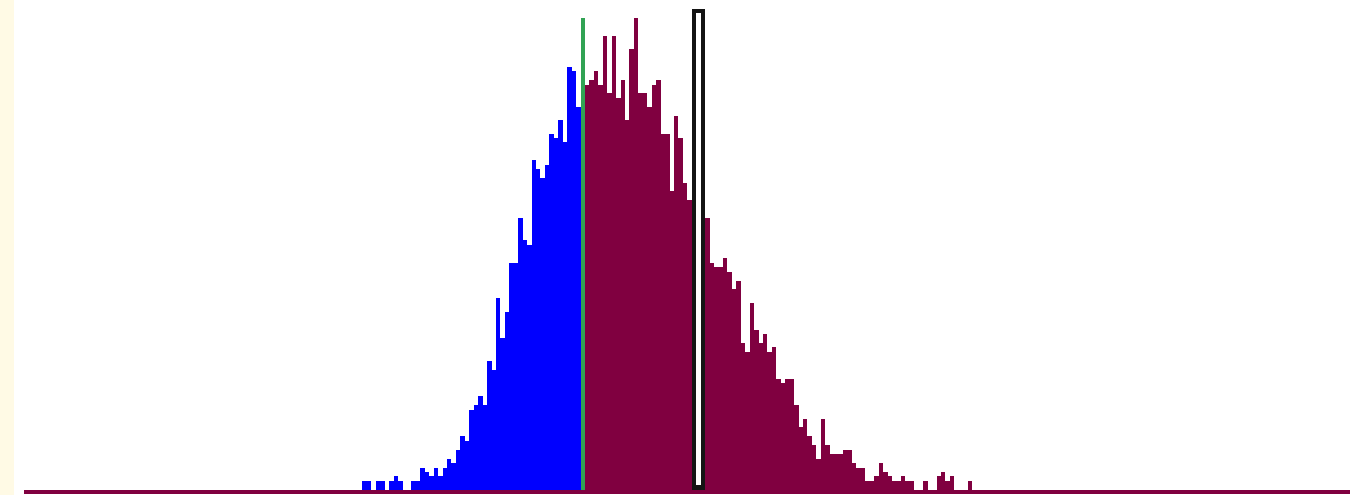


The map shows Casablanca-Settat as a region with high health-related poverty.

Moran's I - Negative Spatial Autocorrelation of Health Deprivation (E_SANTE).



permutations: 10000
pseudo p-value: 0,312469



I: -0,1653 E[I]: -0,0909 mean: -0,0890 sd: 0,1415 z-value: -0,5388

- The permutation test with a p-value of 0.31 rejects Moran's I's statistical significance, suggesting a random spatial pattern of added value.

- The Moran's I value of -0.165 shows negative spatial correlation, with Casablanca-Settat having high added value compared to its neighbors.

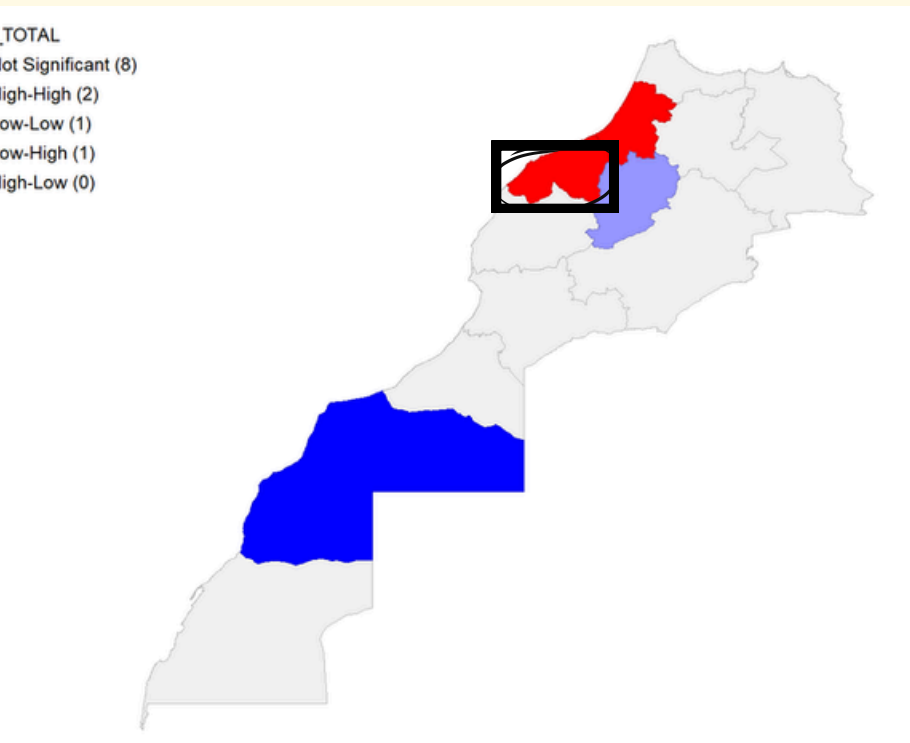
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I. PRESENTATION OF CASABLANCA IN RELATION TO OTHER REGIONS

O SOCIAL LEVEL

EMPLOYMENT

LISA of total employment in Morroco's Regions



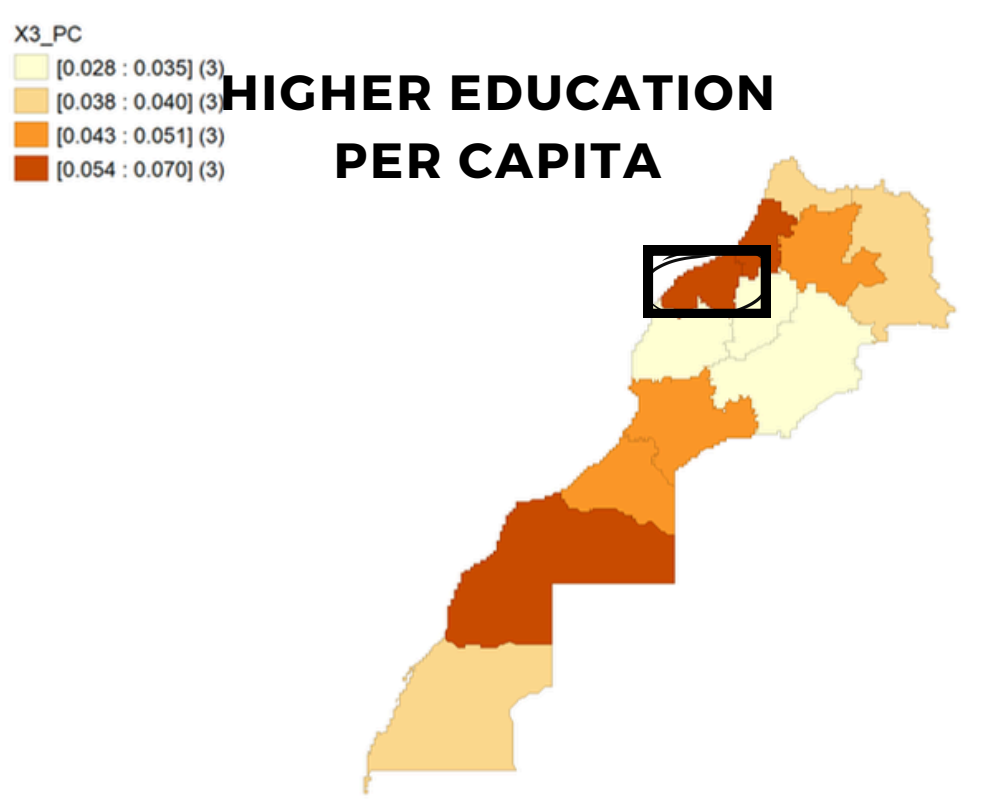
- According to this map, Casablanca-Settat belongs to the High-High cluster, indicating that it, like its neighboring provinces, has a high total number of jobs.
- This high concentration of jobs in Casablanca-Settat can be attributed to its role as a national economic hub, attracting companies and workers alike

I. PRESENTATION OF CASABLANCA IN RELATION TO OTHER REGIONS

O SOCIAL LEVEL

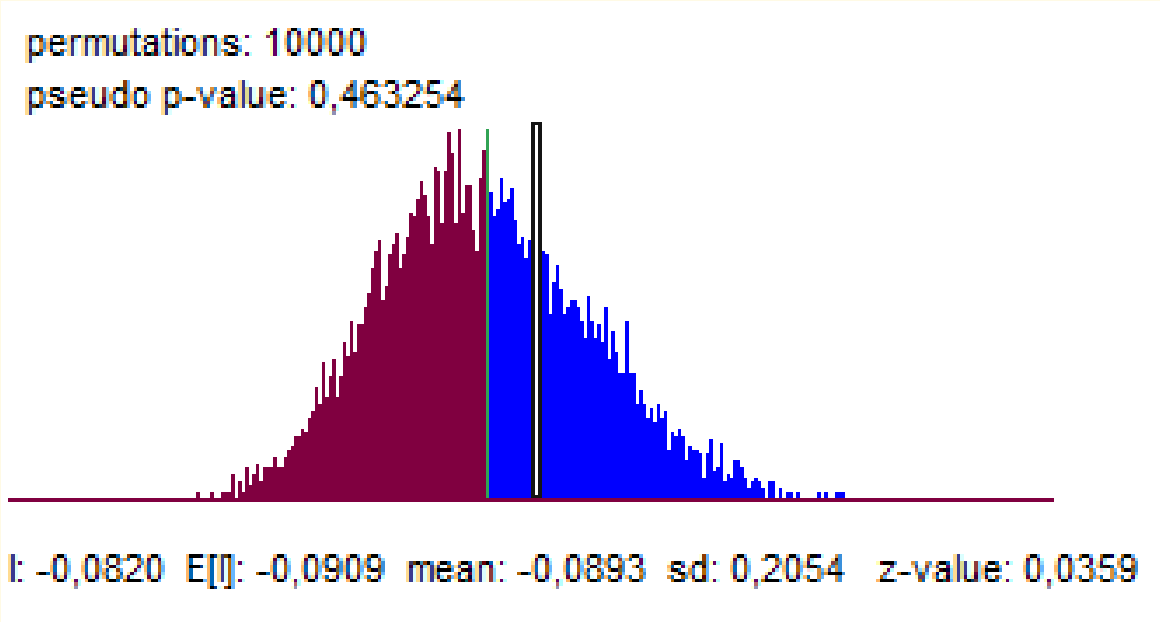
EDUCATION

Geographical Distribution of Higher Education per Capita (X3_PC) in Morocco



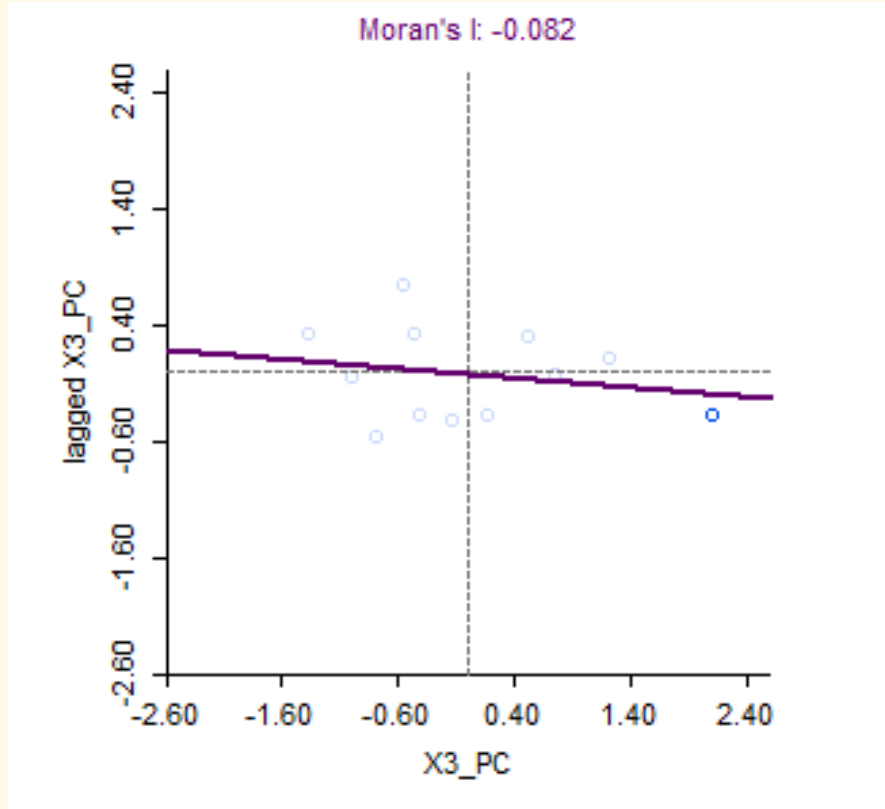
- The map highlights regional disparities in higher education, with northern and central regions showing better access.
- Lighter areas indicate lower access, reflecting education inequalities across Morocco.

Permutation Test for Moran's I - Statistical Significance of Higher Education per Capita (X3_PC)



The permutation test for Moran's I shows that the spatial distribution of higher education per capita is not statistically significant, as indicated by the p-value of 0.463.

Moran's I - Spatial Autocorrelation of Higher Education per Capita (X3_PC)



The Moran's I value of -0.002 indicates that higher education per capita is randomly distributed across regions.

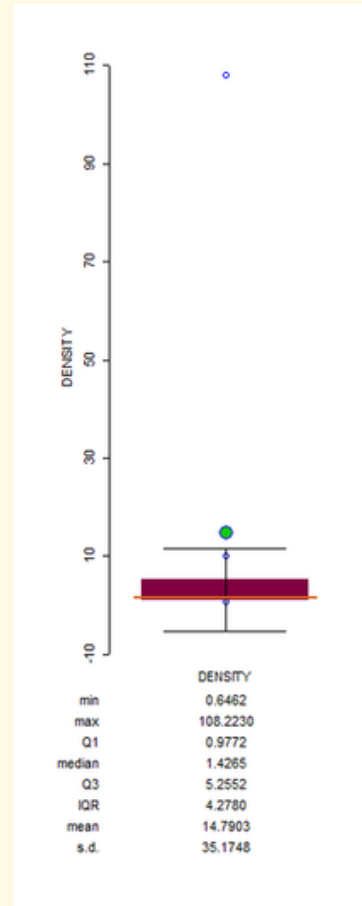
II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION



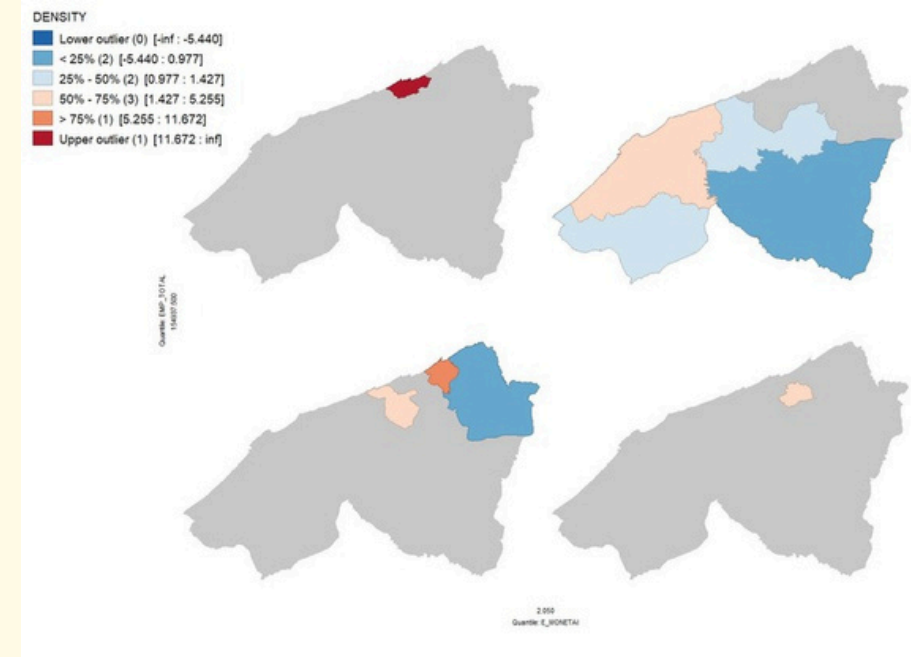
II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

DEMOGRAPHY

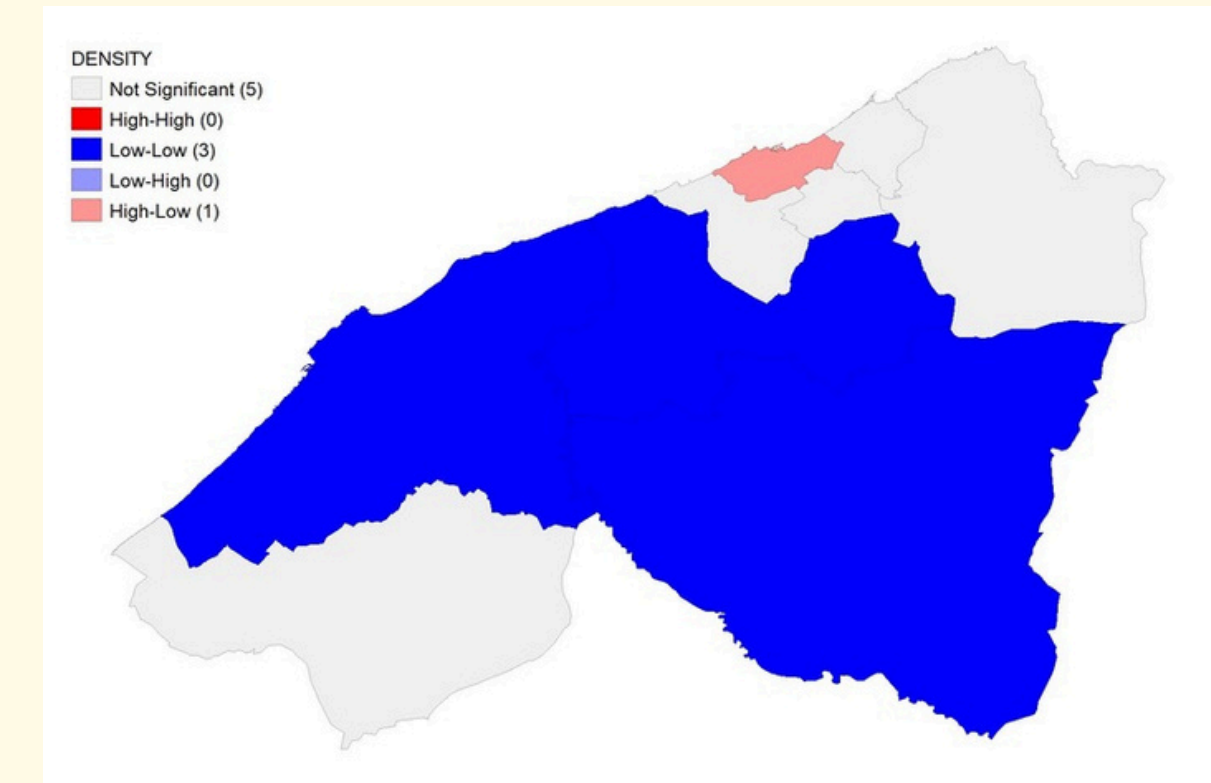
Box Plot and Statistical Summary of Density Distribution



- The box plot shows high variability in density, with significant differences across regions.
- Casablanca province is an outlier, compared to other provinces



Spatial Distribution of Density Significance Across Regions

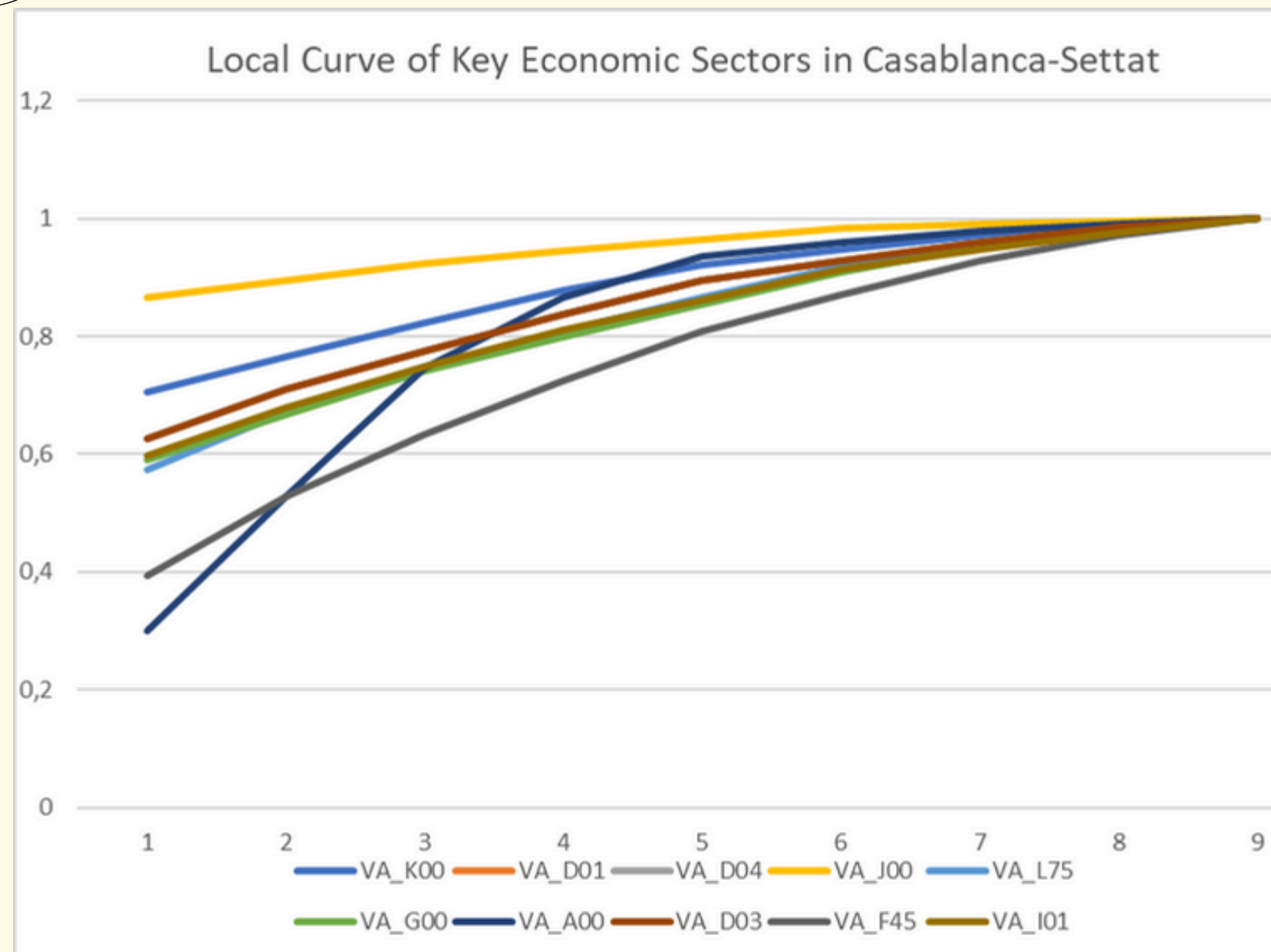


- The map shows that most regions have no significant density, with a small area marked as high-low significance.

- The map displays regional density variations, classified by quartiles and outliers.
- The conditional map showed factors that could explain this disparity.
- According to the conditional map, provinces with high income poverty and low employment are low density, while those with high employment are very dense.
- Monetary poverty and employment could therefore be explanatory factors.

II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

ECONOMY



Source: Excel file Database variable_72PROV_Excel

VA_K00: Immobilier, location et serv. rendus entreprises
VA_D01: Industries alimentaires et tabac
VA_D04: Industrie mécanique, métallurgique et électrique
VA_J00: Activités financières et assurances
VA_L75 Administration publique et sécurité sociale
VA_G00: Commerce
VA_A00: Agriculture, forêt et services annexes
VA_D03: Industrie chimique et parachimique
VA_F45: Bâtiment et travaux publics
VA_I01: Transports

- Local Curves are based on an analysis of the cumulative contribution of the provinces to the value added of the economic sectors.
- The Local Curve of Casablanca-Settat highlights the varying contributions of sectors, with real estate and commerce dominating, while agriculture, transportation, and industrial sectors show a smaller but steady role in regional economic diversification.

II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

ECONOMY

The share of key economic sectors in the Casablanca region.

PROVINCE	VA_A00	VA_D01	VA_D03	VA_D04	VA_F45	VA_G00	VA_I01	VA_J00	VA_K00	VA_L75
Benslimane	0,179	0,070	0,034	0,045	0,085	0,058	0,054	0,021	0,146	0,160
Berrechid	0,134	0,137	0,066	0,088	0,074	0,069	0,060	0,028	0,117	0,075
Casablanca	0,003	0,133	0,064	0,085	0,034	0,071	0,050	0,112	0,178	0,073
El Jadida	0,226	0,096	0,047	0,062	0,080	0,064	0,047	0,026	0,097	0,086
Médiouna	0,028	0,183	0,089	0,117	0,066	0,089	0,053	0,013	0,117	0,061
Mohammadia	0,021	0,168	0,082	0,108	0,053	0,066	0,051	0,036	0,139	0,074
Nouaceur	0,016	0,155	0,075	0,099	0,099	0,065	0,054	0,031	0,194	0,044
Settat	0,265	0,072	0,035	0,046	0,097	0,095	0,045	0,028	0,069	0,107
Sidi Bennour	0,360	0,087	0,042	0,055	0,075	0,099	0,045	0,011	0,043	0,070
Casablanca-Settat	0,065	0,127	0,062	0,082	0,052	0,072	0,050	0,078	0,152	0,077

- This table highlights the economic heterogeneity of the Casablanca-Settat provinces, with marked sectoral specialization.
- Casablanca dominates in services and finance, while provinces such as Settat and El Jadida remain firmly rooted in agriculture.
- Industry is well represented in areas such as Berrechid and Mohammedia, confirming the region's economic diversity.

Source: Excel file Database variable_72PROV_Excel

II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

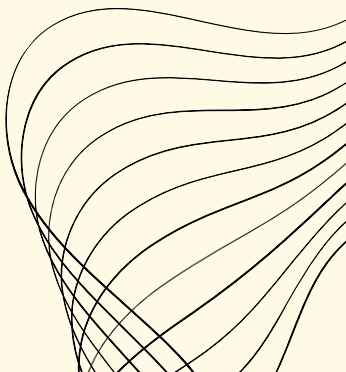
ECONOMY

Provincial share

PROVINCE	VA_A00	VA_D01	VA_D03	VA_D04	VA_F45	VA_G00	VA_I01	VA_J00	VA_K00	VA_L75	VA_TOTAL
Benslimane	0,070	0,014	0,014	0,014	0,042	0,020	0,027	0,007	0,024	0,053	0,014
Berrechid	0,118	0,062	0,062	0,062	0,083	0,055	0,069	0,021	0,044	0,056	0,062
Casablanca	0,024	0,627	0,627	0,627	0,394	0,591	0,598	0,865	0,705	0,574	0,627
El Jadida	0,301	0,066	0,066	0,066	0,134	0,077	0,081	0,029	0,055	0,097	0,066
Médiouna	0,010	0,033	0,033	0,033	0,029	0,028	0,024	0,004	0,018	0,018	0,033
Mohammadia	0,020	0,082	0,082	0,082	0,064	0,057	0,063	0,029	0,057	0,060	0,082
Nouaceur	0,012	0,058	0,058	0,058	0,091	0,043	0,052	0,019	0,061	0,028	0,058
Settat	0,228	0,032	0,032	0,032	0,105	0,074	0,051	0,020	0,026	0,078	0,032
Sidi Bennour	0,218	0,027	0,027	0,027	0,058	0,054	0,036	0,006	0,011	0,036	0,027
Casablanca-Settat	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000

- Casablanca-Settat is an economically polarized region, with Casablanca as the main driving force, particularly in services and industry.
- Some provinces, however, retain strong specializations (agriculture in El Jadida and Settat, industry in Mohammedia and Berrechid).
- Agricultural provinces have low population density, vast farmland, and high value-added, while El Jadida's coastal location favors strong fishing activity.

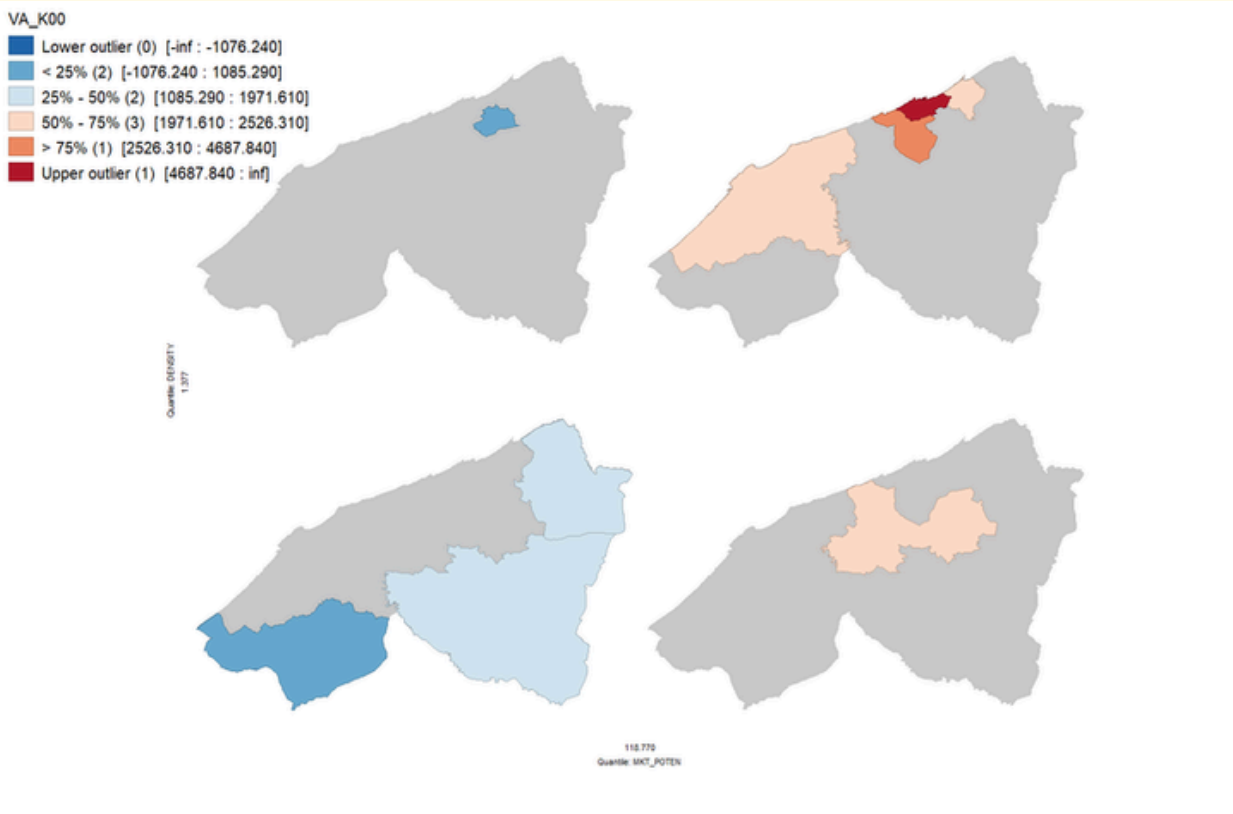
Source: Excel file Database variable_72PROV_Excel



II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

ECONOMY

The share of key economic sectors in the Casablanca region.



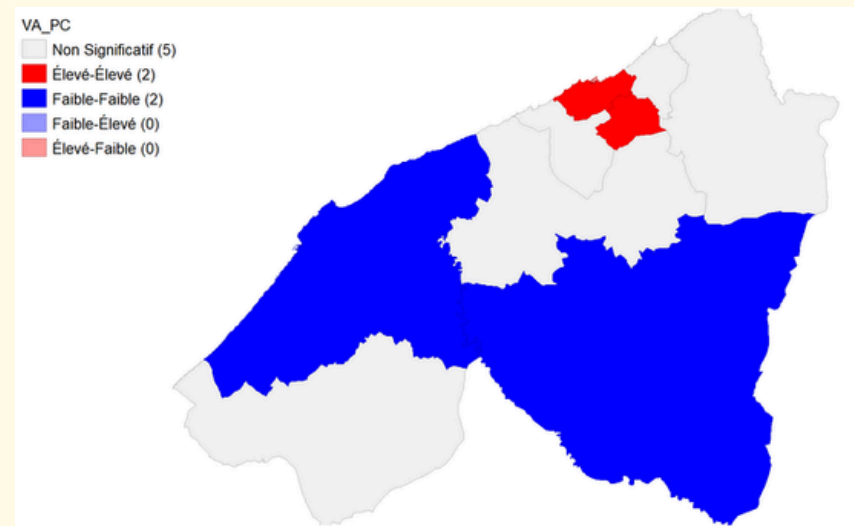
- Provinces with higher real estate value (VA_K00) also show higher market potential (MKT_POTEN).
- Densely populated and economically active areas drive investment through increased demand for housing and infrastructure.
- Casablanca and nearby regions exemplify this trend, boosting the regional economy.

Source: Excel file Database variable_72PROV_Excel

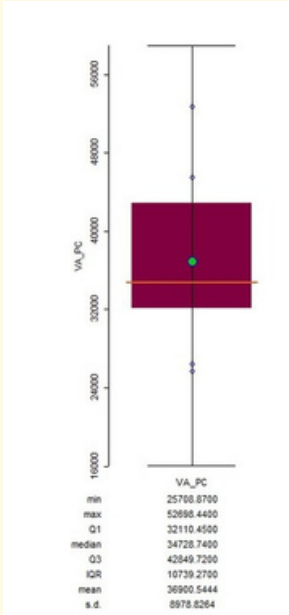
II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

ECONOMY

Spatial Distribution of Added Value per Capita (VA_PC) - Significance and Classification



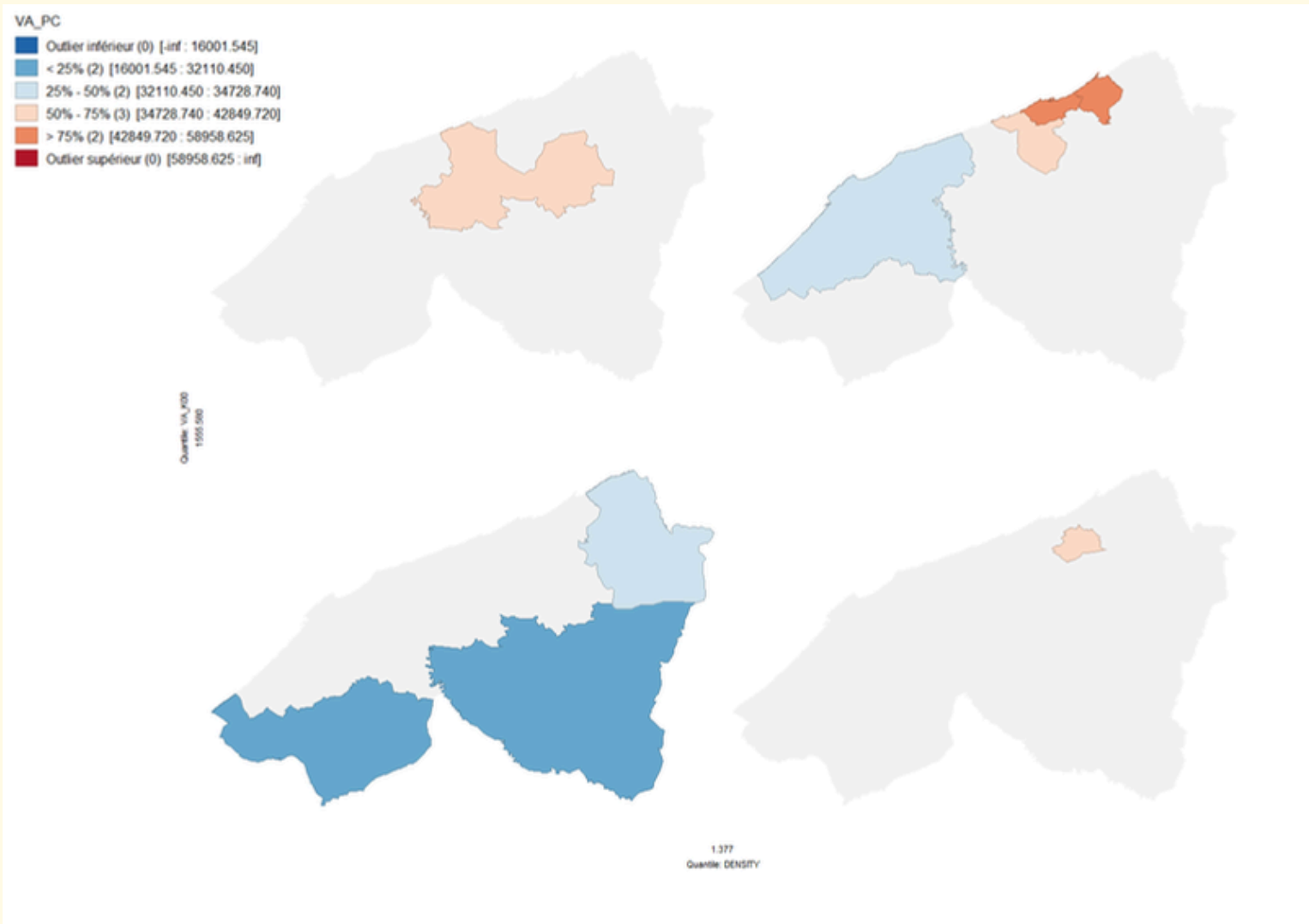
The map illustrates the spatial distribution of added value per capita (VA_PC), showing areas of high (red) economic significance low value (blue), , and regions without significant variation (gray).



Box Plot and Statistical Summary of Added Value per Capita (VA_PC).

The box plot shows a high variability in added value per capita (VA_PC), with a median value of 34,278.70 and a wide interquartile range, indicating significant economic differences across regions.

Spatial Distribution of Added Value per Capita (VA_PC) by Quartile and Outlier Classification



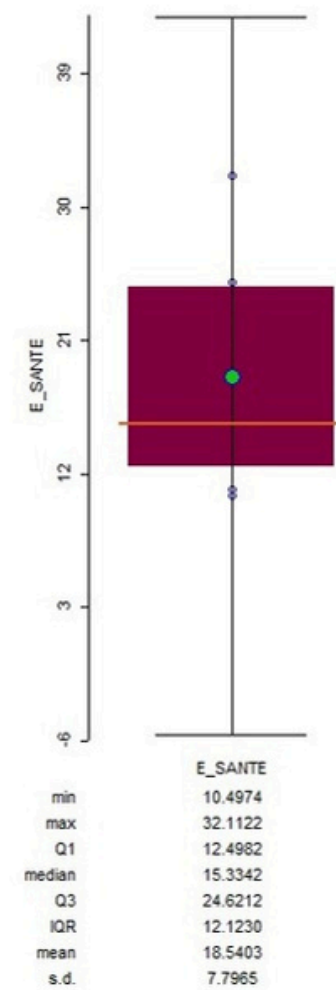
The map shows the spatial distribution of added value per capita (VA_PC), classified by quartiles and outliers, highlighting significant variations in economic value across regions.

II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

SOCIAL LEVEL

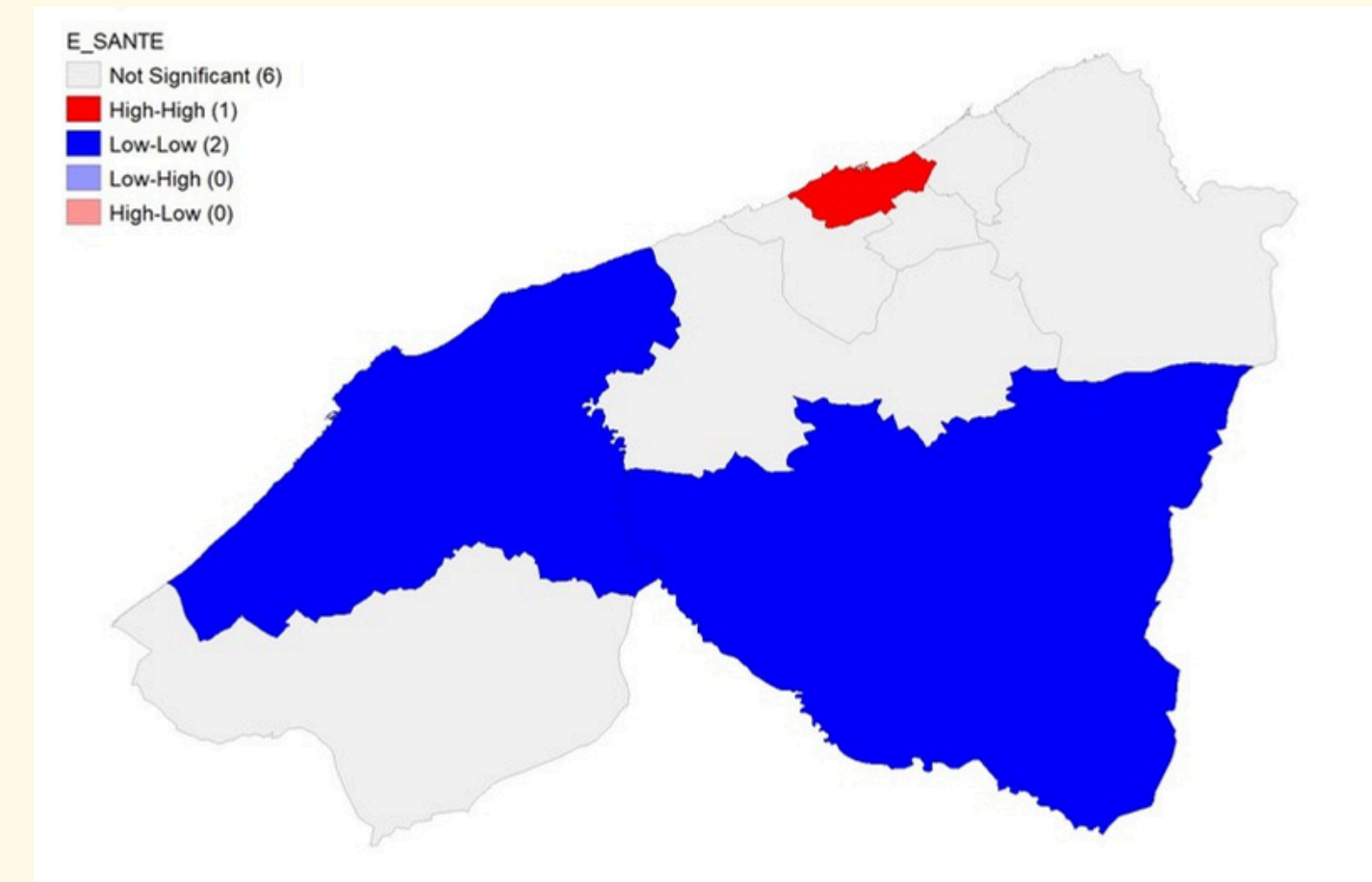
HEALTH

Box Plot and Statistical Summary of Health Deprivation (E_SANTE)



The box plot shows moderate variability in health deprivation (E_SANTE), with a median of 15.34 and an interquartile range indicating a spread of values across regions, suggesting disparities in health access.

Spatial Distribution of Health Deprivation (E_SANTE) - Significance and Classification



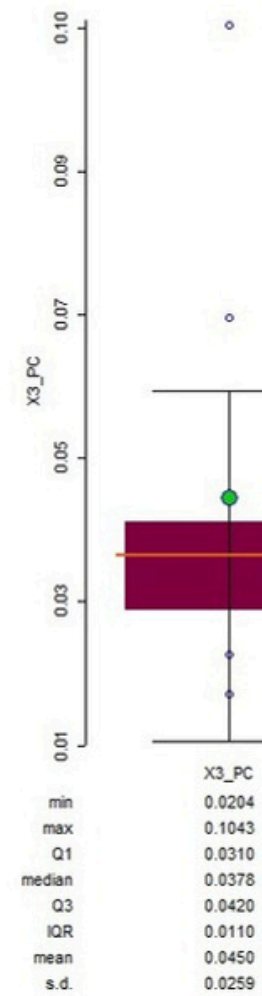
- The map shows the spatial distribution of health deprivation (E_SANTE) in Casablanca-Settat, highlighting regions with high or low deprivation.
- Blue provinces indicate lower health deprivation, while the red region highlights areas with higher health-related poverty.
- This emphasizes the need for focused health interventions and resource allocation in the red region to reduce disparities.

II. INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

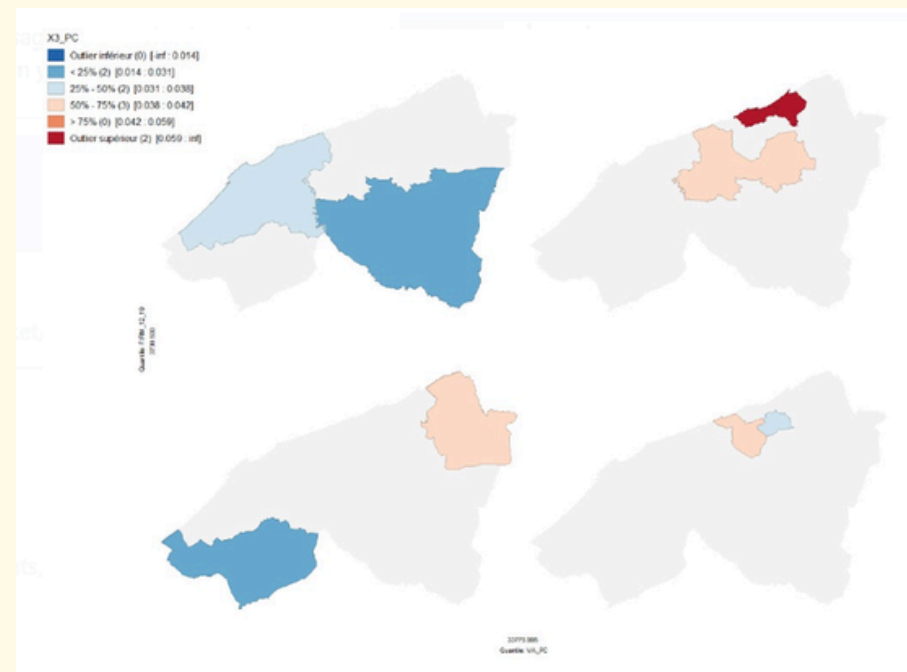
SOCIAL LEVEL

EDUCATION

Box Plot and Statistical Summary of Higher Education per Capita (X3_PC) in Casablanca Provinces

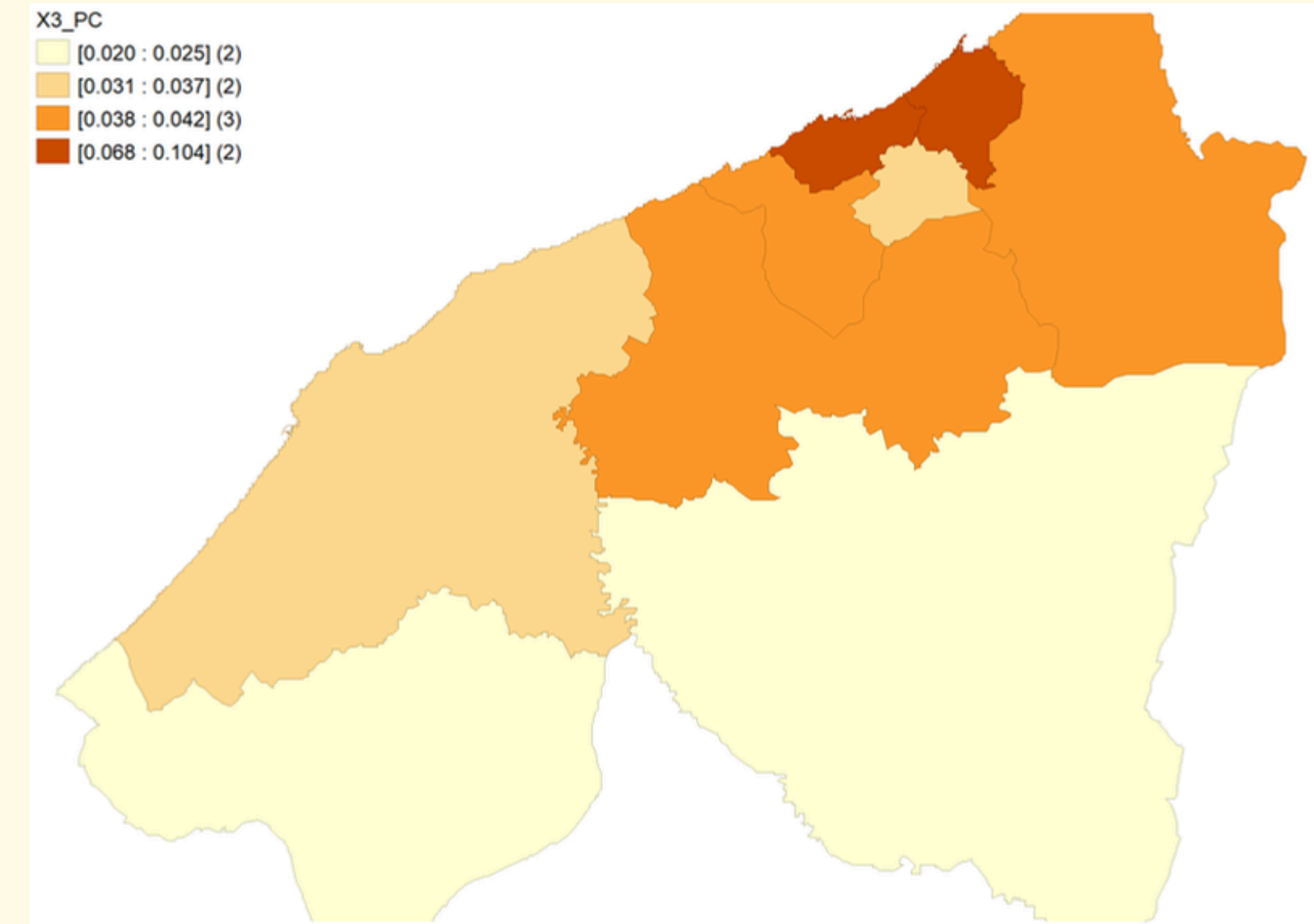


Mohammedia and Casablanca stand out with higher education levels, as shown on this map.



The map shows that higher education access is higher in urbanized, economically developed areas with better infrastructure and resources. Disparities are driven by economic factors like real estate value and market potential.

Geographical Distribution of Higher Education per Capita (X3_PC) in Casablanca Provinces



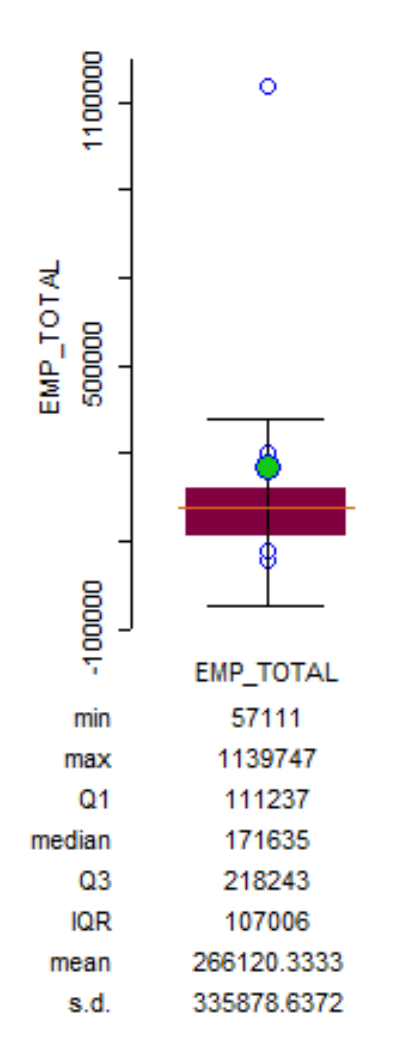
- Higher education is more concentrated in major cities
- Rural provinces have fewer educational infrastructures.
- Limited access can hinder economic opportunities.

II.INTRA-REGIONAL ANALYSIS OF THE CASABLANCA-SETTAT REGION

SOCIAL LEVEL

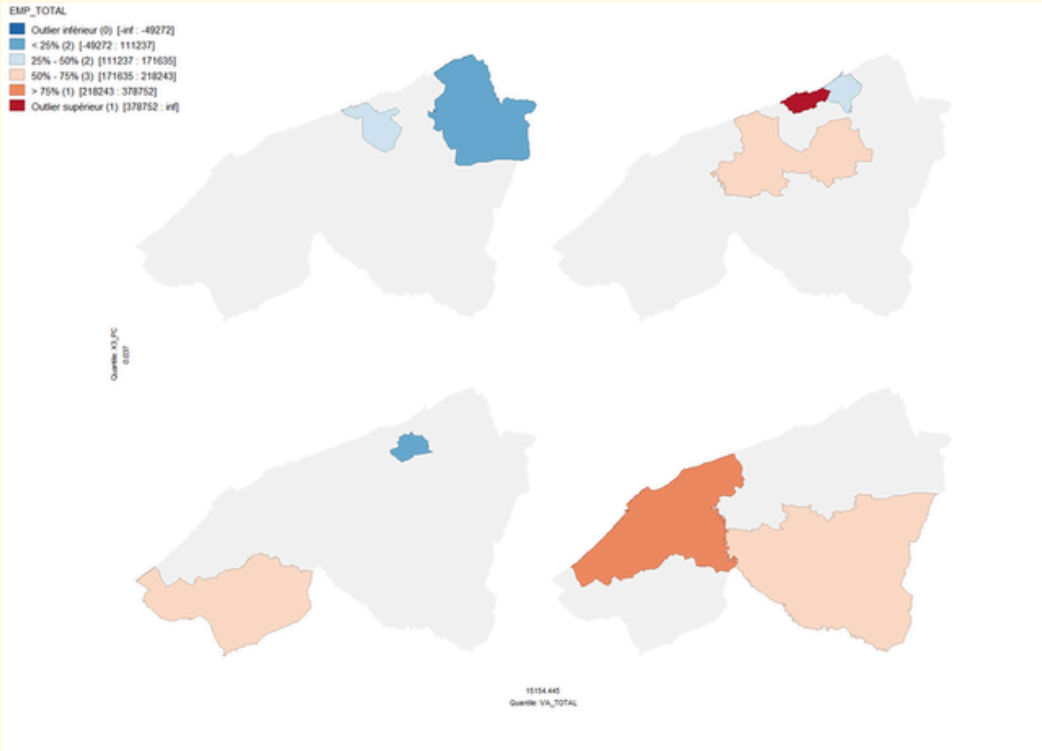
EMPLOYEMENT

Box Plot and Statistical Summary of Total Employment (EMP_TOTAL) in Casablanca Provinces

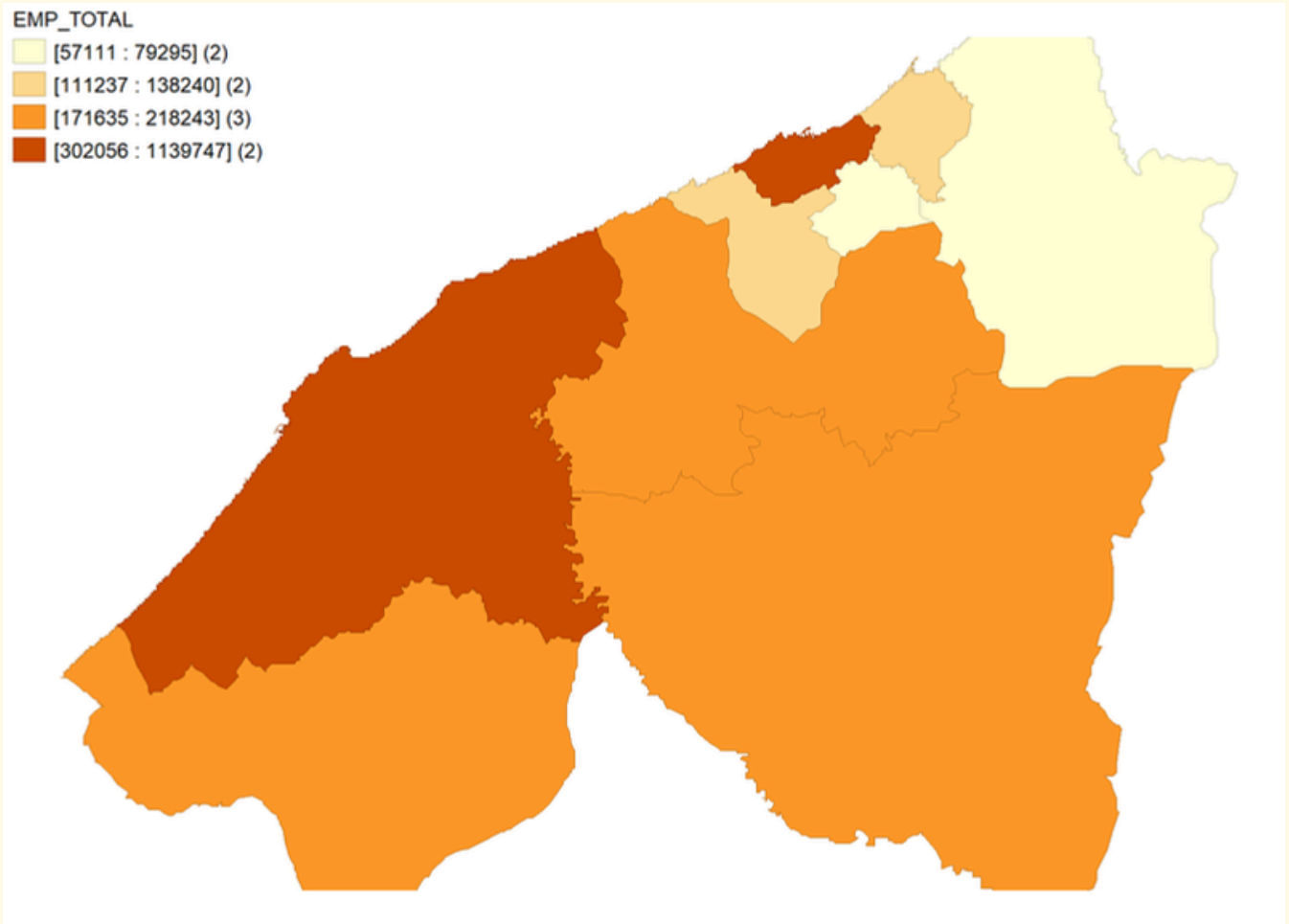


Casablanca is the sole employment outlier despite El Jadida also showing high employment.

Conditional Map of Employment between total value added and higher education



Geographical Distribution of Total Employment (EMP_TOTAL) in Casablanca Provinces



- Some provinces have significantly higher or lower employment levels.
- Some regions have strong economic activity, while others face employment shortages.
- We observe a concentration of jobs in certain provinces, which can influence economic development and urban planning policies

RECOMMENDATIONS

Economic dimension

Casablanca: Enhance real estate, sustainable construction, and diversify into manufacturing and finance.

Benslimane and Berrechid: Develop transport and trade infrastructure, and invest in education and training.

Mohammadia and Nouaceur: Upgrade industry, energy, encourage renewables, and create economic zones.

El Jadida, Settat, and rural areas: Focus on sustainable agriculture, energy transition, and infrastructure.

Regional approach: Balance investment to reduce urban-rural gaps.

Demographic dimension

Low-density regions:

Invest in infrastructure, support agriculture and local industries, and improve education and healthcare.

High-density region:

Focus on sustainable urban planning, reduce infrastructure pressure, and diversify the economy.

Balanced strategy:

Prioritize investments in low-density areas while supporting urban growth with tailored policies.

RECOMMENDATIONS

Social dimension

Health Deprivation :

- Enhance healthcare in high deprivation areas and launch preventive health campaigns in less deprived regions.

Higher Education :

- Invest in rural education infrastructure and encourage private sector innovation in urban areas.

Employment :

- Promote job creation in underdeveloped areas, offer vocational training, and invest in urban infrastructure to boost employment.
- Prioritize development in underdeveloped regions while supporting growth in prosperous areas.

CONCLUSION

In conclusion, the analysis of the Casablanca-Settat provinces reveals disparities in health, education, and employment between urban and rural areas. Targeted policies are necessary to improve access to healthcare services, expand education in rural regions, and promote job creation. An integrated regional approach, balancing investments, is essential to reduce inequalities and improve the quality of life.

THANK YOU

